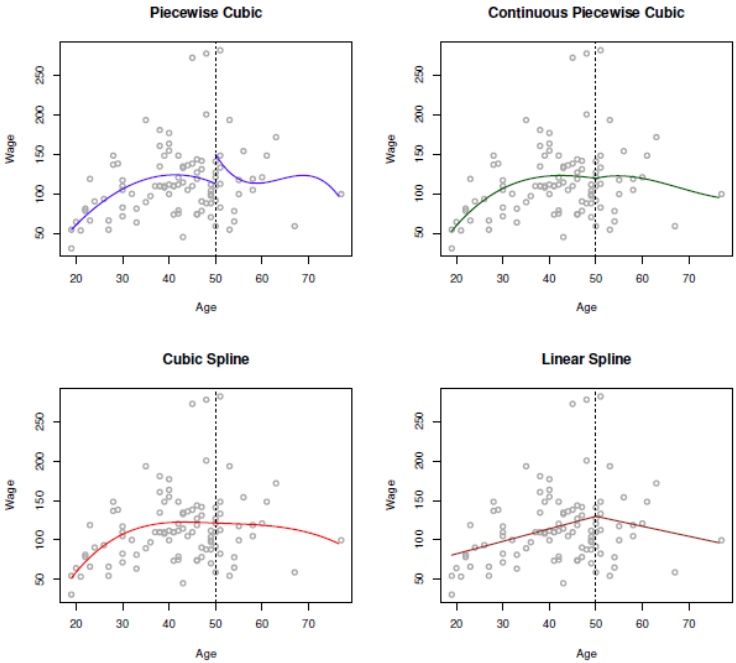
7.3 Basis Functions

Piecewise Polynomials

- Instead of a single polynomial in X over its whole domain, w can use different polynomials defined by knots.
- Ex:

$$y_i = \begin{cases} \beta_{01} + \beta_{11}x_i + \beta 21 x_i^2 + \beta_{31}x_i^3 + \epsilon_i & \text{if} \quad x_i < c \\ \beta_{02} + \beta_{12}x_i + \beta 22 x_i^2 + \beta_{32}x_i^3 + \epsilon_i & \text{if} \quad x_i \geq c \end{cases}$$

- Better for adding constraints to the polynomials (e.g., continuity)
  - *Splines* have the maximum amount of continuity


Piecewise Cubic
Continuous Piecewise Cubic
Cubic Spline
Linear Spline
Wage
Age
50
100
150
200
250
20
30
40
50
60
70

Linear Splines

A linear spline with knots at ξ_k, $k = 1, \ldots, K$ is a piecewise linear polynomial that is continuous at each knot.

We can represent this as

$$y_i = \beta_0 + \beta_1 b_1(x_i) + \beta_2 b_2(x_i) + \cdots + \beta_{K+1} b_{K+1}(x_i) + \epsilon_i$$

where the b_k are what we call *basis functions*.

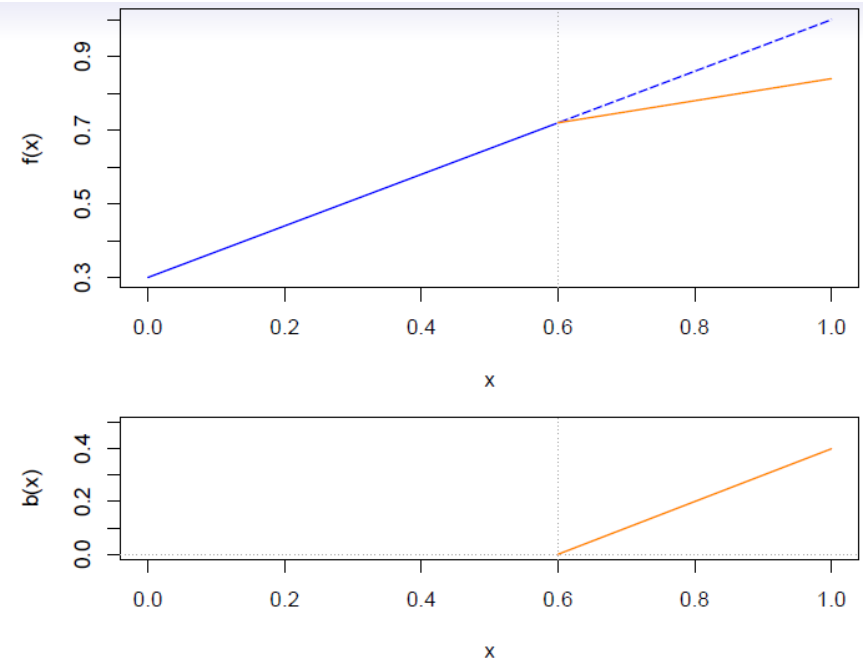
Basis Functions

We define basis functions b_k as

$$
\begin{aligned}
b_1(x_i) &= x_i \\
b_{k+1}(x_i) &= (x_i - \xi_k)_+, \quad k = 1, \ldots, L
\end{aligned}
$$

where $(x)_+$ means the *positive part* of x:

$$
(x)_+ = \begin{cases} x_i & \text{if} \quad x > 0 \\ 0 & \text{otherwise} \end{cases}
$$


f(x)
0.3
0.5
0.7
0.9
0.0
0.2
0.4
0.6
0.8
1.0
x
b(x)
0.0
0.2
0.4
0.0
0.2
0.4
0.6
0.8
1.0
x

Cubic Splines

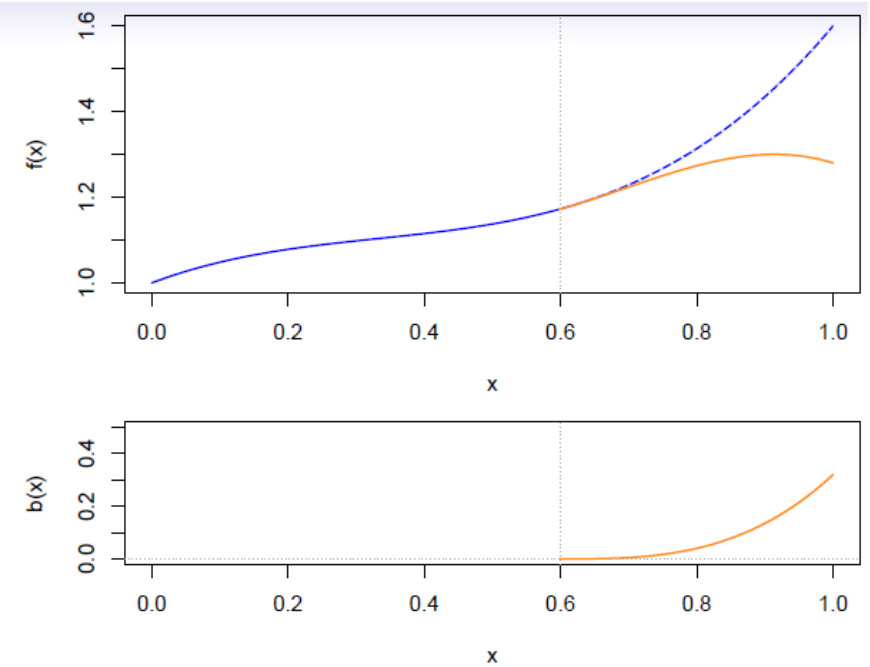
A cubic spline with knots at ξ_k, $k = 1, \ldots, K$ is a piecewise cubic polynomial with continuous derivatives up to order 2 at each knot.

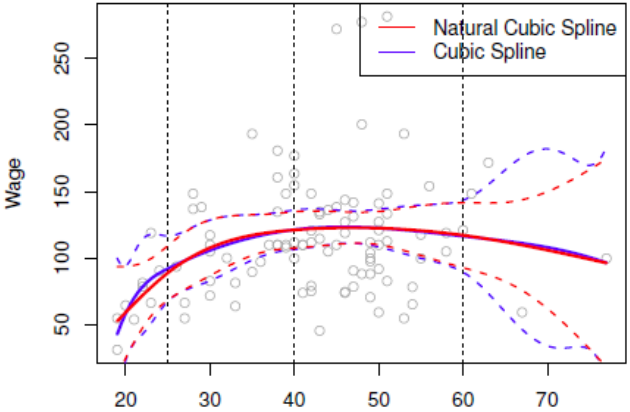
We can represent this as

$$y_i = \beta_0 + \beta_1 b_1(x_i) + \beta_2 b_2(x_i) + \cdots + \beta_{K+3} b_{K+3}(x_i) + \epsilon_i$$

where now the basis functions go through

$$b_{k+3}(x_i) = (x_i - \xi_k)_+^3, \quad k = 1, \ldots, K$$


f(x)
1.0
1.2
1.4
1.6
0.0
0.2
0.4
0.6
0.8
1.0
x
b(x)
0.0
0.2
0.4
0.0
0.2
0.4
0.6
0.8
1.0
x

Natural Cubic Splines

- A natural cubic spline extrapolate linearly beyond the boundary knots.
- This adds $4 = 2 \times 2$ additional constraints.
- Allows us to put more internal knows for the same degrees of freedom as a regular cubic spline.